THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $4x + 3y = 31$ and $3x + 4y = 25$, then $x + y =$ 1

(a) 56 **(b)** 6 **(c)** 8 **(d)** 28

2. If $x + y = 11$ and $x - y = 5$, then $x^2 - y^2 =$ 1

(a) 55 **(b)** 146 **(c)** 96 **(d)** 16

3. The value of x in $6x + 5y = 60$, $7x + 4y = 59$ is: 1

(a) -5 **(b)** 11 **(c)** 5 **(d)** 6

4. The value of y in $9x + 7y = 69$, $4x + 5y = 42$ is: 1

(a) -6 **(b)** 6 **(c)** 9 **(d)** 3

5. The pair $4x + ky = 12$, $8x + 8y = 24$ has infinitely many solutions when $k =$ 1

(a) 8 **(b)** 5 **(c)** 4 **(d)** 12

6. For the pair $kx + 4y = 7$ and $9x + 3y = 2$ to have a unique solution, k must NOT equal: 1

(a) $\frac{27}{4}$ **(b)** 12 **(c)** 13 **(d)** $\frac{63}{2}$

7. If $5x + 2y = 33$ and $2x + 5y = 27$, then $x + y =$ 1

(a) 2 **(b)** 30 **(c)** 60 **(d)** $\frac{60}{7}$

8. If $x + y = 13$ and $x - y = 4$, then $x^2 - y^2 =$ 1

(a) 185 **(b)** 17 **(c)** 52 **(d)** 153

9. The value of x in $9x + 2y = 50$, $5x + 3y = 41$ is: 1

(a) 7 **(b)** 4 **(c)** -4 **(d)** 11

10. The value of y in $12x + 5y = 70$, $7x + 8y = 51$ is: 1

(a) -2 **(b)** 5 **(c)** 2 **(d)** 7

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. For what value of k does the pair $kx + 3y = 4$, $8x + 2y = 5$ have a unique solution? 2

12. Find k so that the pair $3x + ky = 9$ and $6x + 6y = 18$ has infinitely many solutions. 2

13. Solve by the cross-multiplication method: $6x + 5y = 60$ and $7x + 4y = 59$. 2

14. For what value of k does the pair $kx + 4y = 7$, $9x + 3y = 2$ have a unique solution? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve by elimination: $6x + 5y = 60$ and $7x + 4y = 59$. **3**

16. The sum of the digits of a two-digit number is 9. If the digits are interchanged, the number becomes 45 more. Find the number. **3**

17. Find the value of k for which the pair $kx + 3y = k - 3$ and $12x + ky = k$ has infinitely many solutions. **3**

OR

Solve by elimination: $9x + 2y = 50$ and $5x + 3y = 41$. **3**

18. The sum of the digits of a two-digit number is 7. If the digits are interchanged, the number becomes 45 more. Find the number. **3**

OR

Find the value of k for which the pair $kx + 4y = k - 4$ and $16x + ky = k$ has infinitely many solutions. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A father is 9 times as old as his son. After 12 years the father will be 5 times as old as the son will be. Find their present ages. **5**

OR

A fraction becomes $\frac{9}{13}$ when 2 is added to both the numerator and the denominator, and becomes $\frac{11}{15}$ when 4 is added to both. Find the fraction. **5**

20. The taxi charges consist of a fixed charge plus a charge per km. For 5 km the fare is ₹70 and for 12 km it is ₹140. Find the fixed charge, the charge per km and the fare for 20 km. **5**

OR

If the length of a rectangle is reduced by 3 units and its breadth is increased by 2 units, its area decreases by 6 square units. If the length is increased by 2 units and the breadth by 2 units, the area increases by 54 square units. Find the length and breadth. **5**

END OF QUESTION PAPER — Total: 40 Marks

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